

CHAPTER THREE

KINETICS OF A PARTICLE: FORCE AND ACCELERATION

Learning Objectives

After completing this chapter, students should be able to do the following:

- Analyse the accelerated motion of a particle using the equation of motion with different coordinate systems.

3.1 Newton's Second Law of Motion

Kinetics is a branch of dynamics that deals with the relationship between the change in motion of a body and the forces that cause this change. The basis for kinetics is Newton's second law, which states that when an unbalanced force acts on a particle, the particle will accelerate in the direction of the force with a magnitude that is proportional to the force.

If the mass of the particle is m , Newton's second law of motion may be written in mathematical form as

$$\mathbf{F} = m\mathbf{a} \quad (3.1)$$

The above equation, which is referred to as the equation of motion, is one of the most important formulations in mechanics.

3.2 Newton's Law of Gravitational Attraction

Shortly after formulating his three laws of motion, Newton postulated a law governing the mutual attraction between any two particles. In mathematical form this law can be expressed as

$$F = G \frac{m_1 m_2}{r^2} \quad (3.2)$$

Where,

F = force of attraction between the two particles

G = universal constant of gravitation; according to experimental evidence,

$$G = 66.73(10^{-12}) \text{ m}^3/(\text{kg}\cdot\text{s}^2) \text{ or } 66.73 \times 10^{-12} \text{ Nm}^3/\text{kg}^2$$

m_1, m_2 = mass of each of the two particles

r = distance between the centers of the two particles

3.3 Equation of Motion for a System of Particles

Considering the instant, the arbitrary i -th particle, having a mass m_i , is subjected to a system of a resultant external force. The resultant external force F_i represents, for example, the effect of gravitational, electrical, magnetic, or contact forces between the i th particle and adjacent

bodies or particles not included within the system. The equation of motion, written for the system of particles, becomes

$$\sum \mathbf{F}_i = \sum m_i \mathbf{a}_i \quad (3.3)$$

If $\mathbf{r}_G$ is a position vector which locates the center of mass G of the particles, then by definition of the center of mass:

$$m\mathbf{r}_G = \sum m_i \mathbf{r}_i \quad (3.4)$$

$$\text{where } m = \sum m_i \quad (3.5)$$

is the total mass of all the particles.

Differentiating this equation twice with respect to time, assuming that no mass is entering or leaving the system, yields

$$m\mathbf{a}_G = \sum m_i \mathbf{a}_i \quad (3.6)$$

Substituting this result into Eq. 3.3, we obtain

$$\sum \mathbf{F} = m\mathbf{a}_G \quad (3.7)$$

Hence, the sum of the external forces acting on the system of particles is equal to the total mass of the particles times the acceleration of its center of mass G. Since in reality all particles must have a finite size to possess mass, Eq. 3.7 justifies application of the equation of motion to a body that is represented as a single particle.

3.4 Equations of Motion: Rectangular Coordinates

When a particle moves relative to an inertial x, y, z frame of reference, the forces acting on the particle, as well as its acceleration, can be expressed in terms of their $\mathbf{i}, \mathbf{j}, \mathbf{k}$ components, Fig. 3.1.

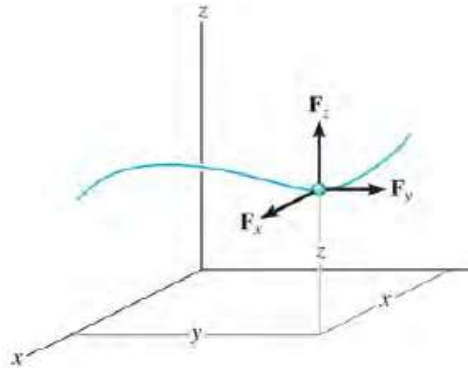


Fig. 3.1

Applying the equation of motion ($\sum \mathbf{F} = m\mathbf{a}$), we have:

$$\mathbf{F} = F_x\mathbf{i} + F_y\mathbf{j} + F_z\mathbf{k} = m(a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{k}) \quad (3.8)$$

For this equation to be satisfied, the respective $\mathbf{i}, \mathbf{j}, \mathbf{k}$ components on the left side must equal the corresponding components on the right side. Consequently, we may write the following three scalar equations:

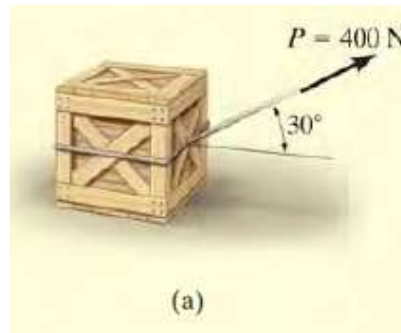
$$\begin{aligned} \sum F_x &= ma_x \\ \sum F_y &= ma_y \\ \sum F_z &= ma_z \end{aligned} \quad (3.9)$$

In particular, if the particle is constrained to move only in the x-y plane, then the first two of these equations are used to specify the motion.

Example 3.1

The 50-kg crate shown in Fig. E-3.1a rests on a horizontal surface for which the coefficient of kinetic friction is $\mu_k = 0.3$. If the crate is subjected to a 400-N towing force as shown,

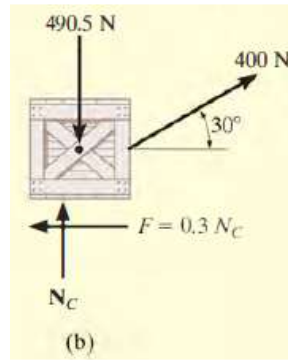
(i) determine the velocity of the crate in 3 s starting from rest, (ii) Find the position of the crate after the time interval.



Solution

Using the equations of motion, we can relate the crate's acceleration to the force causing the motion. The crate's velocity can then be determined using kinematics.

Free-Body Diagram. The weight of the crate is $W = mg = 50 \text{ kg} (9.81 \text{ m/s}^2) = 490.5 \text{ N}$. As shown in Fig. E-3.1b, the frictional force has a magnitude $F = \mu_k N_C$ and acts to the left, since it opposes the motion of the crate. The acceleration a is assumed to act horizontally, in the positive x direction. There are two unknowns, namely N_C and a .



Equations of Motion. Using the data shown on the free-body diagram, we have

$$\rightarrow \Sigma F_x = ma_x; \quad 400 \cos 30^\circ - 0.3N_C = 50a \quad (1)$$

$$+\uparrow \Sigma F_y = ma_y; \quad N_C - 490.5 + 400 \sin 30^\circ = 0 \quad (2)$$

Solving Eq. 2 for N_C , substituting the result into Eq. 1, and solving for a yields

$$N_C = 290.5 \text{ N}$$

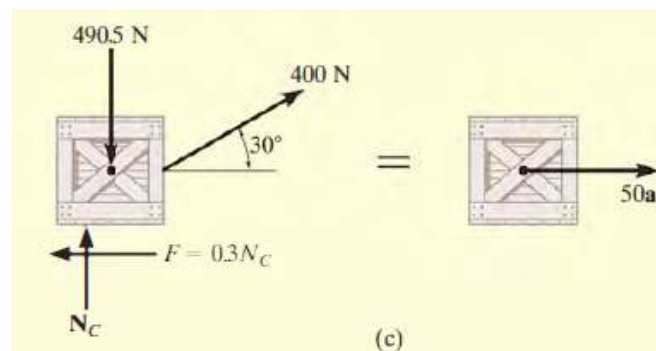
$$a = 5.185 \text{ m/s}^2$$

Kinematics. Notice that the acceleration is *constant*, since the applied force $\mathbf{P}$ is constant. Since the initial velocity is zero, the velocity of the crate in 3 s is

$$\begin{aligned} (\rightarrow) \quad v &= v_0 + a_c t = 0 + 5.185(3) \\ &= 15.6 \text{ m/s} \rightarrow \end{aligned}$$

Ans.

Note: We can also use the alternative procedure of drawing the crate's free-body and kinetic diagrams, Fig. E-3.1c, prior to applying the equations of motion.



Example 3.2

A 10-kg projectile is fired vertically upward from the ground, with an initial velocity of 50 m/s, Fig. E- 3.2a. Determine the maximum height to which it will travel if (a) atmospheric resistance is neglected; and (b) atmospheric resistance is measured as $F_D = (0.01v^2)$ N, where v is the speed of the projectile at any instant, measured in m/s.

SOLUTION

In both cases the known force on the projectile can be related to its acceleration using the equation of motion. Kinematics can then be used to relate the projectile's acceleration to its position.

Part (a) Free-Body Diagram. As shown in Fig. E- 3.2b, the projectile's weight is $W = mg = 10(9.81) = 98.1$ N. We will assume the unknown acceleration a acts upward in the *positive* z direction.

Equation of Motion.

$$+\uparrow \Sigma F_z = ma_z; \quad -98.1 = 10a, \quad a = -9.81 \text{ m/s}^2$$

The result indicates that the projectile, like every object having free-flight motion near the earth's surface, is subjected to a *constant* downward acceleration of 9.81 m/s^2 .

Kinematics. Initially, $z_0 = 0$ and $v_0 = 50 \text{ m/s}$, and at the maximum height $z = h$, $v = 0$. Since the acceleration is *constant*, then

$$\begin{aligned} (+\uparrow) \quad v^2 &= v_0^2 + 2a_c(z - z_0) \\ 0 &= (50)^2 + 2(-9.81)(h - 0) \\ h &= 127 \text{ m} \end{aligned} \quad \text{Ans}$$

Part (b) Free-Body Diagram. Since the force $F_D = (0.01v^2)$ N tends to retard the upward motion of the projectile, it acts downward as shown on the free-body diagram, Fig. E- 3.2c.

Equation of Motion.

$$+\uparrow \Sigma F_z = ma_z; \quad -0.01v^2 - 98.1 = 10a, \quad a = -(0.001v^2 + 9.81)$$

Kinematics. Here the acceleration is *not constant* since F_D depends on the velocity. Since $a = f(v)$, we can relate a to position using

$$(+\uparrow) a \, dz = v \, dv; \quad -(0.001v^2 + 9.81) \, dz = v \, dv$$

Separating the variables and integrating, realizing that initially $z_0 = 0$, $v_0 = 50 \text{ m/s}$ (positive upward), and at $z = h$, $v = 0$, we have

$$\begin{aligned} \int_0^h dz &= - \int_{50}^0 \frac{v \, dv}{0.001v^2 + 9.81} = -500 \ln(v^2 + 9810) \Big|_{50 \text{ m/s}}^0 \\ h &= 114 \text{ m} \end{aligned} \quad \text{Ans.}$$

NOTE: The answer indicates a lower elevation than that obtained in part (a) due to atmospheric resistance or drag.

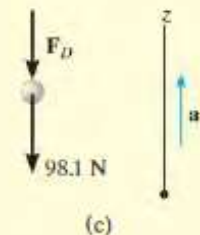
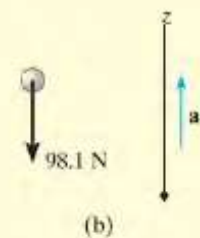
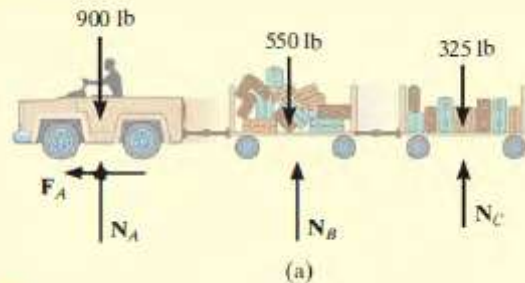


Fig. E- 3.2

Example 3.3



The baggage truck *A* shown in the photo has a weight of 900 lb and tows a 550-lb cart *B* and a 325-lb cart *C*. For a short time the driving frictional force developed at the wheels of the truck is $F_A = (40t)$ lb, where t is in seconds. If the truck starts from rest, determine its speed in 2 seconds. Also, what is the horizontal force acting on the coupling between the truck and cart *B* at this instant? Neglect the size of the truck and carts.



SOLUTION

Free-Body Diagram. As shown in Fig. E- 3.3a, it is the frictional driving force that gives both the truck and carts an acceleration. Here we have considered all three vehicles as a single system.

Equation of Motion. Only motion in the horizontal direction has to be considered.

$$\begin{aligned} \pm \Sigma F_x &= ma_x; & 40t &= \left(\frac{900 + 550 + 325}{32.2} \right) a \\ & & a &= 0.7256t \end{aligned}$$

Kinematics. Since the acceleration is a function of time, the velocity of the truck is obtained using $a = dv/dt$ with the initial condition that $v_0 = 0$ at $t = 0$. We have

$$\int_0^v dv = \int_0^{2s} 0.7256t \, dt, \quad v = 0.3628t^2 \Big|_0^{2s} = 1.45 \text{ ft/s} \quad \text{Ans.}$$

Free-Body Diagram. In order to determine the force between the truck and cart *B*, we will consider a free-body diagram of the truck so that we can “expose” the coupling force *T* as external to the free-body diagram, Fig. E- 3.3b.

Equation of Motion. When $t = 2$ s, then

$$\begin{aligned} \pm \Sigma F_x &= ma_x; & 40(2) - T &= \left(\frac{900}{32.2} \right) [0.7256(2)] \\ & & T &= 39.4 \text{ lb} \quad \text{Ans.} \end{aligned}$$

NOTE: Try and obtain this same result by considering a free-body diagram of carts *B* and *C* as a single system.

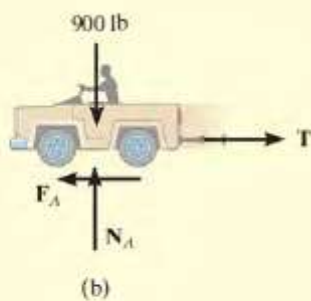


Fig. E- 3.3

Example 3.4

A smooth 2-kg collar C , shown in Fig. E- 3.4a, is attached to a spring having a stiffness $k = 3 \text{ N/m}$ and an unstretched length of 0.75 m . If the collar is released from rest at A , determine its acceleration and the normal force of the rod on the collar at the instant $y = 1 \text{ m}$.

SOLUTION

Free-Body Diagram. The free-body diagram of the collar when it is located at the arbitrary position y is shown in Fig. E- 3.4b. Furthermore, the collar is *assumed* to be accelerating so that “ a ” acts downward in the *positive* y direction. There are four unknowns, namely, N_C , F_s , a , and θ .

Equations of Motion.

$$\rightarrow \Sigma F_x = ma_x; \quad -N_C + F_s \cos \theta = 0 \quad (1)$$

$$+\downarrow \Sigma F_y = ma_y; \quad 19.62 - F_s \sin \theta = 2a \quad (2)$$

From Eq. 2 it is seen that the acceleration depends on the magnitude and direction of the spring force. Solution for N_C and a is possible once F_s and θ are known.

The magnitude of the spring force is a function of the stretch s of the spring; i.e., $F_s = ks$. Here the unstretched length is $AB = 0.75 \text{ m}$, Fig. E- 3.4a; therefore, $s = CB - AB = \sqrt{y^2 + (0.75)^2} - 0.75$. Since $k = 3 \text{ N/m}$, then

$$F_s = ks = 3(\sqrt{y^2 + (0.75)^2} - 0.75) \quad (3)$$

From Fig. E- 3.4a, the angle θ is related to y by trigonometry.

$$\tan \theta = \frac{y}{0.75} \quad (4)$$

Substituting $y = 1 \text{ m}$ into Eqs. 3 and 4 yields $F_s = 1.50 \text{ N}$ and $\theta = 53.1^\circ$. Substituting these results into Eqs. 1 and 2, we obtain

$$N_C = 0.900 \text{ N} \quad \text{Ans}$$

$$a = 9.21 \text{ m/s}^2 \downarrow \quad \text{Ans}$$

NOTE: This is not a case of constant acceleration, since the spring force changes both its magnitude and direction as the collar moves downward.

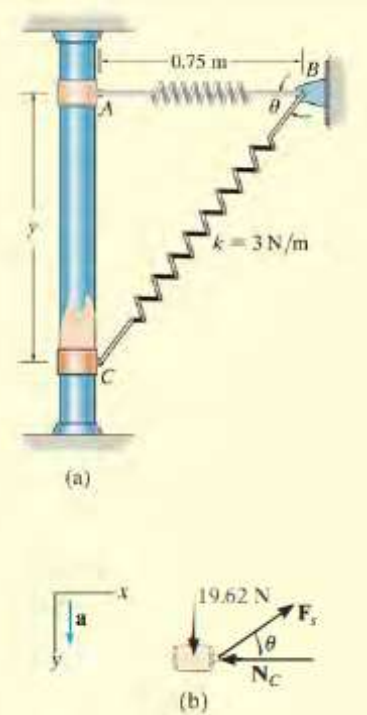


Fig. E- 3.4

Example 3.5

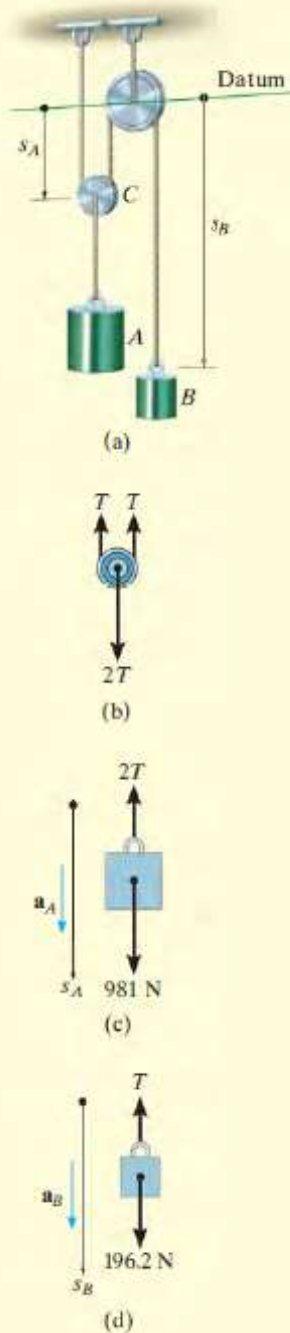


Fig. E- 3.5

The 100-kg block A shown in Fig. E- 3.5a is released from rest. If the masses of the pulleys and the cord are neglected, determine the speed of the 20-kg block B in 2 s.

SOLUTION

Free-Body Diagrams. Since the mass of the pulleys is neglected, then for pulley C , $ma = 0$ and we can apply $\Sigma F_y = 0$ as shown in Fig. E- 3.5b. The free-body diagrams for blocks A and B are shown in Fig. E- 3.5c and d , respectively. Notice that for A to remain stationary $T = 490.5 \text{ N}$, whereas for B to remain static $T = 196.2 \text{ N}$. Hence A will move down while B moves up. Although this is the case, we will assume both blocks accelerate downward, in the direction of $+s_A$ and $+s_B$. The three unknowns are T , a_A , and a_B .

Equations of Motion. Block A ,

$$+\downarrow \Sigma F_y = ma_y; \quad 981 - 2T = 100a_A \quad (1)$$

Block B ,

$$+\downarrow \Sigma F_y = ma_y; \quad 196.2 - T = 20a_B \quad (2)$$

Kinematics. The necessary third equation is obtained by relating a_A to a_B using a dependent motion analysis, discussed earlier. The coordinates s_A and s_B in Fig. E- 3.5a measure the positions of A and B from the fixed datum. It is seen that

$$2s_A + s_B = l$$

where l is constant and represents the total vertical length of cord. Differentiating this expression twice with respect to time yields

$$2a_A = -a_B \quad (3)$$

Notice that when writing Eqs. 1 to 3, the *positive direction was always assumed downward*. It is very important to be *consistent* in this assumption since we are seeking a simultaneous solution of equations. The results are

$$T = 327.0 \text{ N}$$

$$a_A = 3.27 \text{ m/s}^2$$

$$a_B = -6.54 \text{ m/s}^2$$

Hence when block A accelerates *downward*, block B accelerates *upward* as expected. Since a_B is constant, the velocity of block B in 2 s is thus

$$(+\downarrow)$$

$$\begin{aligned} v &= v_0 + a_B t \\ &= 0 + (-6.54)(2) \\ &= -13.1 \text{ m/s} \end{aligned}$$

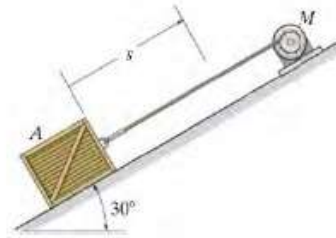
Ans.

The negative sign indicates that block B is moving upward.

Problem Set 3.1

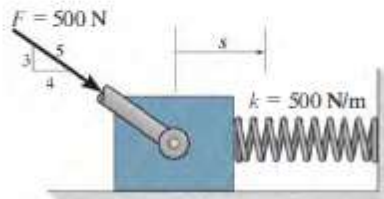
1.

The motor winds in the cable with a constant acceleration, such that the 20-kg crate moves a distance $s = 6$ m in 3 s, starting from rest. Determine the tension developed in the cable. The coefficient of kinetic friction between the crate and the plane is $\mu_k = 0.3$.



2.

A spring of stiffness $k = 500$ N/m is mounted against the 10-kg block. If the block is subjected to the force of $F = 500$ N, determine its velocity at $s = 0.5$ m. When $s = 0$, the block is at rest and the spring is uncompressed. The contact surface is smooth.



3.

The casting has a mass of 3 Mg. Suspended in a vertical position and initially at rest, it is given an upward speed of 200 mm/s in 0.3 s using a crane hook H . Determine the tension in cables AC and AB during this time interval if the acceleration is constant.

